

Highlights

- In-region cross-validation overstates crop-map accuracy and misranks models.
- Model rankings invert between in-region and spatially disjoint holdout tests.
- Sparse, axis-aligned feature selection, not deep capacity, drives spatial transfer.
- A sparse-attention encoder (L-TAE-S) ties TabNet for best transfer, trains faster.
- Field-level aggregation is a second, substitutable lever for robust transfer.